# FORMAL SMOOTHNESS IS UNCHANGED BY PASSING TO COMPLETIONS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

*Editorial note.* Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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*Editorial note.* Benign-screen note: the proof contains the phrase 'it is clear that' / 'is obvious' once, used as a legitimate step inside a complete argument; the author proves the statement, so this is not a hint-only proof.

*Allowed standard prerequisites (editorial).*

- Adic completeness of the completion, agreement of the extended ideal, and the quotients by powers of the ideal. Algebra, Lemma \ref{algebra-lemma-hathat-finitely-generated} and its proof, cited at proof lines 9242-9247. Identify the lifting diagrams of the three formulations.
- The definition of formal smoothness through lifting diagrams. Definition \ref{definition-formally-smooth}, cited at proof lines 9248-9250. Is the property compared across the three formulations.

*Source reference.* Original lemma 07ED: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9443–9443. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

*Source reference.* Original lemma 05GG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7060–7060. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

*Source reference.* Original lemma 07EB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9441–9441. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

**Lemma 0.2.** *Let $(R, \mathfrak{m})$ and $(S, \mathfrak{n})$ be rings endowed with finitely generated ideals. Endow $R$ and $S$ with the $\mathfrak{m}$-adic and $\mathfrak{n}$-adic topologies. Let $R \to S$ be a homomorphism of topological rings. The following are equivalent*

(1) *$R \to S$ is formally smooth for the $\mathfrak{n}$-adic topology,*
(2) *$R \to S^\wedge$ is formally smooth for the $\mathfrak{n}^\wedge$-adic topology,*
(3) *$R^\wedge \to S^\wedge$ is formally smooth for the $\mathfrak{n}^\wedge$-adic topology.*

*Here $R^\wedge$ and $S^\wedge$ are the $\mathfrak{m}$-adic and $\mathfrak{n}$-adic completions of $R$ and $S$.*

*Proof.* The assumption that $\mathfrak{m}$ is finitely generated implies that $R^\wedge$ is $\mathfrak{m}R^\wedge$-adically complete, that $\mathfrak{m}R^\wedge = \mathfrak{m}^\wedge$ and that $R^\wedge/\mathfrak{m}^n R^\wedge = R/\mathfrak{m}^n$, see Algebra, Lemma 05GG and its proof. Similarly for $(S, \mathfrak{n})$. Thus it is clear that diagrams as in Definition 07EB for the cases (1), (2), and (3) are in 1-to-1 correspondence. □